

LORS-Enhanced Graph Transformers Framework: Accelerating Convergence and Enriching Representation with Oscillatory Activation

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Abstract. Graph Transformers (GTs) leverage global attention to model long-range interactions in graphs and often outperform message-passing Graph Neural Networks (GNNs). Yet many GTs converge slowly and struggle with dynamic or nonlinear structural patterns, limiting their practicality on time- and budget-constrained tasks. We propose LORS-GT, a Graph Transformer framework that augments attention and activations with Lee Oscillators with Retrograde Signaling (LORS), a biologically inspired oscillatory module that modulates features dynamically without adding inference overhead. To the best of our knowledge, LORS-GT is the first GT framework with oscillatory activation mechanisms. We instantiate LORS-GT on two state-of-the-art GT backbones and several representative GNN architectures, and evaluate it on 14 GT benchmarks and 1 GNN benchmark spanning homophilic and heterophilic graphs. Across these datasets, LORS consistently improves accuracy and accelerates early-stage convergence over the corresponding baselines, especially under low-epoch training budgets.

Keywords: Graph Transformers · Lee Oscillators · Oscillatory Signaling
· Graph Neural Networks · Transient-Chaotic Dynamics

1 INTRODUCTION

Graph-structured data arise in social networks, biochemical molecule analysis, and recommendation systems, where message-passing Graph Neural Networks (GNNs) suffer from limited receptive fields and over-smoothing or over-squashing, weakening their ability to capture long-range structural dependencies [14]. Graph Transformers (GTs) extend attention to graphs, enabling global interactions between node pairs and offering higher theoretical expressivity [8].

Despite this progress, GTs still face practical limitations: slow convergence, difficulty modeling dynamic or nonlinear structures, and instability when depth

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or training budget is constrained. These issues reduce their effectiveness in time-sensitive or noisy graph applications.

To address them, we propose LORS-GT, a framework that injects chaotic neural dynamics into Graph Transformers via Lee Oscillators with Retrograde Signaling (LORS) [11]. LORS dynamically modulates attention and activation layers without additional inference cost. Applied to Polynormer [5] and Tokenphormer [20], LORS-GT improves performance on 15 homophilic and heterophilic graph benchmarks and accelerates convergence, especially under limited training budgets.

Our main technical contributions are as follows:

- We are the first to integrate Lee Oscillators with Retrograde Signaling (LORS) into Graph Transformer architectures, introducing biologically inspired oscillatory dynamics for dynamic feature modulation beyond static attention.
- We design a lightweight integration scheme that injects LORS-based chaotic signals into attention and activation layers, improving expressivity and optimization.
- Instantiated on Polynormer and Tokenphormer, LORS-GT achieves state-of-the-art or competitive performance on 11 of 14 graph benchmarks, consistently improving accuracy and convergence speed.
- We also extend LORS to several traditional GNNs, where it yields substantial gains, demonstrating its generality beyond Transformer-based models.

2 RELATED WORK

2.1 Graph Transformers: Expressivity and Oversmoothing

Graph Transformers (GTs) adapt attention-based architectures from NLP and vision to graph data. Early models incorporate structural priors such as shortest-path or positional encodings into self-attention, but still incur quadratic complexity in the number of nodes [8]. More recent designs improve scalability and structure-awareness: Polynormer achieves linear-time local-to-global attention with polynomial expressivity, while Tokenphormer augments each node with structure-aware tokens derived from random walks and hop distances, which is useful in low-label regimes [5, 20].

Deep GTs can still suffer from oversmoothing, where node representations homogenize with depth and expressive power decays [14]. Nkansah et al. show that attention-based GNNs, including GATs and GTs, may experience exponential decay in representational capacity due to graph connectivity, motivating mechanisms beyond static attention alone [13]. Existing remedies include residual connections and initial feature reuse, as in GCNII [2], and non-convolutional architectures such as RUM [17]. Our approach is complementary: we inject layerwise oscillatory dynamics into GTs without changing the macro-architecture.

2.2 Oscillatory and Dynamic Mechanisms in Deep Neural Networks

Dynamical-systems perspectives model deep networks as continuous-time systems. Neural Ordinary Differential Equations reinterpret deep networks in this way, and Continuous Graph Neural Networks extend this view to graphs, allowing node features to evolve along learned trajectories while helping mitigate oversmoothing [3, 18].

Related work also explores oscillatory components. Kuramoto-style coupled oscillators model synchronization in interacting units [16], and complex-valued oscillatory neural networks have been proposed for storing and retrieving multi-dimensional aperiodic signals [1].

Our work instead leverages the transient-chaotic Lee Oscillator and its type-2 transient-fuzzy extension CT2TFDNN [10, 9], originally developed for time-series and fuzzy regression tasks. Unlike stable oscillators, these oscillators exhibit controlled bifurcation and transient chaos, helping models escape flat minima while maintaining gradient flow. We are the first to integrate this oscillator family into Graph Transformers, using excitatory-inhibitory feedback to inject nonlinear dynamics into attention and activation components, thereby enhancing convergence and expressivity without changing the macro-architecture.

3 METHODOLOGY

In this section, we introduce the proposed LORS-GT framework, a Graph Transformer architecture that enhances existing GTs by incorporating the LORS function and can be applied to nearly all GT architectures. To demonstrate how LORS improves Graph Transformers, we devise four integration strategies. Section 3.1 introduces the core concepts and structure of LORS, the key module of our framework. Section 3.2 describes the three strategies using two SOTA GTs as examples.

3.1 Structure of LORS

LORS is based on the Lee Oscillator [10] and is characterized by transient-chaotic neural dynamics, enabling it to capture complex nonlinear dependencies and uncertainties in the input data. This makes it well suited for dynamic interactions among nodes in graph data.

Each LORS comprises an excitatory neuron (E), an inhibitory neuron (I), an input neuron (Ω), and an output neuron (LORS). The a_i and b_i are weights in the excitatory and inhibitory neurons, while s_i are the weights of outputs from the excitatory and inhibitory neurons. The parameters d and c are weights in the output neuron, ξ_E and ξ_I are threshold values, and $S(t)$ is the external input stimulus. Their interactions are described by a system of nonlinear equations with a retrograde signaling mechanism, where output signals feed back to the input and inhibitory neurons. This structure improves the model’s ability to

capture dynamic information and mitigates overtraining and deadlock in traditional neural networks [19]. The LORS used in this paper is a lightly improved variant of the original one, with several additional weight variables.

The neural dynamics of LORS are given below:

$$tempE(t) = a_1 LORS(t) + a_2 E(t) - a_3 I(t) + a_4 S(t) - \xi_E \quad (1)$$

$$tempI(t) = b_1 LORS(t) - b_2 E(t) - b_3 I(t) + b_4 S(t) - \xi_I \quad (2)$$

$$E(t+1) = Sig(s_1 * tempE(t)) \quad (3)$$

$$I(t+1) = Sig(s_2 * tempI(t)) \quad (4)$$

$$\Omega(t+1) = Sig[s_3 * S(t)] \quad (5)$$

$$LORS(t) = d * [E(t) - I(t)] e^{-kS^2(t)} + c * \Omega(t) \quad (6)$$

$$Min_{-kS^2(t)} = m \quad (7)$$

As shown in Figure 1, the neural dynamics of LORS are illustrated through unique bifurcation diagrams, which depict alternating stable and chaotic regions. The transient-chaotic zones effectively represent complex nonlinear data variations.

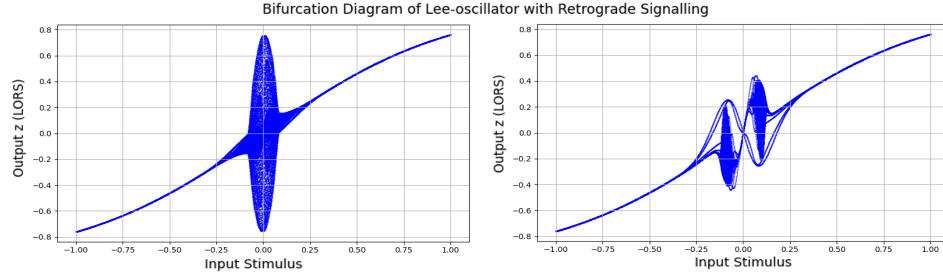


Figure 1. Two Type of Typical LORS

3.2 LORS-GT Integration

Graph Transformers (GTs) capture long-range dependencies and global information propagation in graph data, but their self-attention mechanism can cause oversmoothing and weaken sensitivity to local feature variations. Integrating LORS introduces oscillatory dynamics that make GTs more responsive to spatial feature changes.

The chaotic oscillatory nature of LORS provides an adaptive nonlinear mapping mechanism for modeling complex, high-order relationships. Inspired by retrograde neuronal signaling, LORS uses excitatory-inhibitory interactions to stabilize neuron outputs and prevent uncontrolled activation. This feedback mechanism helps the model capture dynamic node relationships under different network states, improves generalization, and maintains gradient flow by preventing neurons from remaining in flat zero-gradient regions.

Overall, LORS enriches feature representations and improves GT performance under challenging conditions. Four types of LORS-GT integration are illustrated in Figure 2.

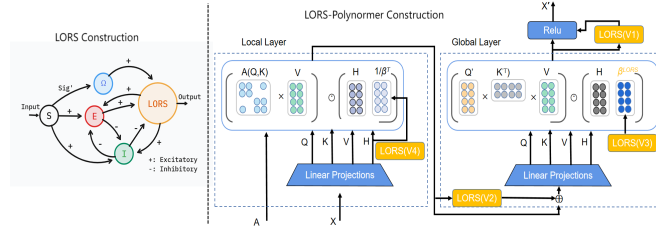


Figure 2. Schematic Diagram of LORS-GT Application in Polynormer

LORS integrate Activation Function with Residual Most activation functions, such as ReLU, provide sparsity by zeroing negative values but remain linear in positive regions, which limits their capacity to model complex nonlinearities. By introducing oscillation dynamics, LORS enhances input data with dynamic nonlinear characteristics, enabling better interaction with high-order system information. Inspired by Residual Networks [7], we partition input data into two paths: one retains the original information, while the other employs LORS for richer feature extraction. The output from LORS then passes through an activation function to maintain information stability. This parallel structure leverages the sparsity advantages of traditional activation functions alongside the dynamic nonlinearity of LORS. The pseudocode is as follows:

Enhance attention output with Residual Graph attention layers aggregate information from neighboring nodes while encoding local structural patterns. Integrating these outputs with LORS introduces dynamic oscillation that strengthens nonlinear dependency modeling. Following the residual design, we add hyperparameters to balance the transformed and original attention, preventing excessive feature distortion. The pseudocode is provided below.

Dynamic feature fusion In Graph Transformers (GTs) and Graph Neural Networks (GNNs), deep architectures may oversmooth node features, reduc-

Algorithm 1: Pseudocode of LORS-Integrated Activation Function (V1)

Input: stimulation x , hyperparameter α, β

- 1 $x \leftarrow \text{lin}(x)$ // Linear transformation;
- 2 $x_LORS \leftarrow \text{LORS}(x)$ // LORS activation;
- 3 $x' \leftarrow \alpha * \text{ReLU}(x) + \beta * \text{ReLU}(x_LORS)$ // Activation and aggregation.
Both ReLU and GELU can be used as activation functions in this integration, here use ReLU as the default activation function;

Algorithm 2: Pseudocode of LORS-Enhanced Attention Output (V2)

Input: attention a , hyperparameter α, β

- 1 $a_LORS \leftarrow \text{LORS}(a)$ // LORS activation;
- 2 $a' \leftarrow \alpha * a + \beta * a_LORS$ // Residual aggregation;

ing discriminative power. Prior GT work addressed this by fusing normalized and original features [5]. LORS-GT further introduces a dynamic feature fusion mechanism. Original features are fed into LORS to produce an oscillatory coefficient x_LORS , which adaptively modulates the fusion ratio between normalized and raw features. Because x_LORS is derived through oscillations, each fused representation becomes uniquely refined. The pseudocode for this process is provided in Algorithm 3 below:

Algorithm 3: Pseudocode of Dynamic Feature Fusion (V3)

Input: stimulation x , hyperparameter α, β

- 1 $x_norm \leftarrow \text{norm}(x)$ // Normalization;
- 2 $x_LORS \leftarrow \text{LORS}(x)$ // Linear transformation;
- 3 $x' \leftarrow (1 - \alpha * x_LORS) * x_norm + \beta * x_LORS * x$ // Feature aggregation;

4 EXPERIMENTS

We integrate four LORS-based strategies—individually or in combination—into two leading Graph Transformer (GT) models, and evaluate the resulting variants against SOTA GNN and GT baselines on nine homophilic and five heterophilic datasets [12, 15]. The tables below report the best performance obtained on each dataset using the optimal integration method. We further incorporate LORS into several conventional GNN baselines and compare the resulting LORS-GNN models with their original versions on the ZINC molecular regression benchmark. Across all evaluations, LORS-enhanced models (LORS-GT and LORS-GNN) consistently outperform their counterparts, demonstrating the effectiveness of LORS in graph representation learning.

We evaluate LORS-Polynormer against more than ten SOTA GNNs and eight advanced GTs on node classification tasks, and compare LORS-Tokenphormer with eight competitive GNNs and ten well-established GTs for node prediction tasks. Performance of baseline GTs is taken from original papers or prior studies, as these results rely on thoroughly tuned implementations. Metrics for the remaining baselines follow previous works [5, 20]. For Polynormer-r and Tokenphormer, we report results based on our own reproductions to ensure consistency when integrating them into LORS-GT.

Table 1. Averaged node classification accuracy (%) $\pm$ std over 10 runs on homophilic datasets. Polynormer-r indicates Polynormer with *ReLU* activation, while LORS-Polynormer denotes the integration of LORS with Polynormer-r. The best results are highlighted in **bold**.

	Year	Computer	Photo	CS	Physics	WikiCS
GCN	2016	89.65 $\pm$ 0.52	92.70 $\pm$ 0.20	92.92 $\pm$ 0.12	96.18 $\pm$ 0.07	77.47 $\pm$ 0.85
GraphSAGE	2017	91.20 $\pm$ 0.29	94.59 $\pm$ 0.14	93.91 $\pm$ 0.13	96.49 $\pm$ 0.06	74.77 $\pm$ 0.95
GAT	2020	90.78 $\pm$ 0.13	93.87 $\pm$ 0.11	93.61 $\pm$ 0.14	96.17 $\pm$ 0.08	76.91 $\pm$ 0.82
GCNII	2020	91.04 $\pm$ 0.41	94.30 $\pm$ 0.20	92.22 $\pm$ 0.14	95.97 $\pm$ 0.11	78.68 $\pm$ 0.55
GGCN	2022	91.81 $\pm$ 0.20	94.50 $\pm$ 0.11	95.25 $\pm$ 0.05	97.07 $\pm$ 0.05	78.44 $\pm$ 0.53
OrderedGNN	2023	92.03 $\pm$ 0.13	95.10 $\pm$ 0.20	95.00 $\pm$ 0.10	97.00 $\pm$ 0.08	79.01 $\pm$ 0.68
tGNN	2022	83.40 $\pm$ 1.33	89.92 $\pm$ 0.72	92.85 $\pm$ 0.48	96.24 $\pm$ 0.24	71.49 $\pm$ 1.05
GraphGPS	2022	91.19 $\pm$ 0.54	95.06 $\pm$ 0.13	93.93 $\pm$ 0.12	97.12 $\pm$ 0.19	78.66 $\pm$ 0.49
NAGphormer	2022	91.22 $\pm$ 0.14	95.49 $\pm$ 0.11	95.75 $\pm$ 0.09	97.34$\pm$0.03	77.16 $\pm$ 0.72
Expormer	2023	91.47 $\pm$ 0.17	95.35 $\pm$ 0.22	94.93 $\pm$ 0.01	96.89 $\pm$ 0.09	78.54 $\pm$ 0.49
NodeFormer	2022	86.98 $\pm$ 0.62	93.46 $\pm$ 0.35	95.64 $\pm$ 0.22	96.45 $\pm$ 0.28	74.73 $\pm$ 0.94
DIFFormer	2023	91.99 $\pm$ 0.76	95.10 $\pm$ 0.47	94.78 $\pm$ 0.20	96.60 $\pm$ 0.18	73.46 $\pm$ 0.56
GOAT	2023	90.96 $\pm$ 0.90	92.96 $\pm$ 1.48	94.21 $\pm$ 0.38	96.24 $\pm$ 0.24	77.00 $\pm$ 0.77
Polynormer	2024	93.18 $\pm$ 0.18	96.11 $\pm$ 0.23	95.51 $\pm$ 0.29	97.22 $\pm$ 0.06	79.53 $\pm$ 0.83
Polynormer-r	2024	93.66 $\pm$ 0.16	96.24 $\pm$ 0.20	95.43 $\pm$ 0.19	97.09 $\pm$ 0.07	79.68 $\pm$ 0.74
LORS-Polynormer Ours		93.77$\pm$0.19	96.50$\pm$0.20	95.86$\pm$0.11	97.21 $\pm$ 0.05	80.31$\pm$0.65

4.1 Effect on Advanced Graph Transformer (Polynormer)

Performance on Homophilic Graph Datasets We first evaluate LORS-Polynormer on 5 homophilic datasets, comparing it with 15 SOTA GNNs and GTs. As shown in Table 1, LORS-Polynormer achieves clear performance improvements after integrating Polynormer-r with LORS, demonstrating the effectiveness of this enhancement within advanced GT architectures. We further observe that although Polynormer typically requires many epochs to reach optimal performance, its results on multiple datasets tend to saturate early. To examine whether LORS accelerates convergence, we evaluate the model after only 10% of the full training epochs. Table 2 shows that the integration of LORS yields even more pronounced improvements in the early training stage across several datasets. The chaotic regime enables fast exploratory behavior and introduces beneficial perturbations near saddle points, helping model rapidly escape local minima [4].

Table 2. Averaged node classification accuracy (%) $\pm$ std over 10 runs on homophilic datasets, with each run completed at 10% of the total epochs. Polynormer-r indicates Polynormer with *ReLU* activation, while LORS-Polynormer denotes the integration of LORS with Polynormer-r. The best results are highlighted in **bold**.

	Year	Computer	Photo	CS	Physics	WikiCS
Polynormer-r	2024	88.89 $\pm$ 0.55	67.48 $\pm$ 14.77	94.94 $\pm$ 0.20	96.80 $\pm$ 0.10	77.41 $\pm$ 0.64
LORS-Polynormer Ours		89.80$\pm$0.57	76.69$\pm$15.58	95.35$\pm$0.22	96.91$\pm$0.09	79.19$\pm$0.83

Performance on Heterophilic Graph Datasets We compare LORS-Polynormer with SOTA GNNs and graph transformers (GTs) on five heterophilic graphs as well. Table 3 reports the mean performance ($\pm$ std) across 10 independent runs, indicating that LORS-Polynormer consistently achieves the overall best performance. Notably, the LORS-enhanced LORS-Polynormer outperforms the original Polynormer across all experiments, further demonstrating the effectiveness of LORS. We also evaluate the performance of Polynormer-r and LORS-Polynormer at 10% of the total training epochs on these five heterophilic datasets, as shown in Table 4. The results further confirm that LORS-driven GTs can effectively accelerate convergence, with accuracy gains of up to 6.57% (from 45.75% to 52.32% on amazon-ratings) achieved by LORS-Polynormer.

Table 3. Averaged node classification results over 10 runs on heterophilic datasets — Accuracy is reported for roman-empire and amazon-ratings, while ROC AUC is reported for minesweeper, tolokers, and questions. Polynormer-r denotes Polynormer with *ReLU* activation. LORS-Polynormer indicates the integration of LORS and Polynormer-r. The best results are highlighted in **bold**.

	Year	roman-empire	amazon-ratings	minesweeper	tolokers	questions
GCN	2016	73.69 $\pm$ 0.74	48.70 $\pm$ 0.63	89.75 $\pm$ 0.52	83.64 $\pm$ 0.67	76.09 $\pm$ 1.27
GraphSAGE	2017	85.74 $\pm$ 0.67	53.63 $\pm$ 0.39	93.51 $\pm$ 0.57	82.43 $\pm$ 0.44	76.44 $\pm$ 0.62
GAT-sep	2020	88.75 $\pm$ 0.41	52.70 $\pm$ 0.62	93.91 $\pm$ 0.35	83.78 $\pm$ 0.43	76.79 $\pm$ 0.71
GGCN+	2022	74.46 $\pm$ 0.54	43.00 $\pm$ 0.32	87.54 $\pm$ 1.22	77.31 $\pm$ 1.14	71.10 $\pm$ 1.57
OrderedGNN	2023	77.68 $\pm$ 0.39	47.29 $\pm$ 0.65	80.58 $\pm$ 1.08	75.60 $\pm$ 1.36	75.09 $\pm$ 1.00
G2NN	2022	82.16 $\pm$ 0.78	47.93 $\pm$ 0.58	91.83 $\pm$ 0.56	82.51 $\pm$ 0.80	74.82 $\pm$ 0.92
DIR-GNN	2023	91.23 $\pm$ 0.32	47.89 $\pm$ 0.39	87.05 $\pm$ 0.69	81.19 $\pm$ 1.05	76.13 $\pm$ 1.24
tGNN	2022	79.95 $\pm$ 0.75	48.21 $\pm$ 0.53	91.93 $\pm$ 0.77	70.84 $\pm$ 1.75	76.38 $\pm$ 1.79
GraphGPS	2022	82.00 $\pm$ 0.61	53.10 $\pm$ 0.42	90.63 $\pm$ 0.67	83.71 $\pm$ 0.48	71.73 $\pm$ 1.47
NAGphormer	2022	74.34 $\pm$ 0.77	51.26 $\pm$ 0.72	84.19 $\pm$ 0.66	78.32 $\pm$ 0.95	68.17 $\pm$ 1.53
Expformer	2023	89.03 $\pm$ 0.37	53.51 $\pm$ 0.46	90.74 $\pm$ 0.53	83.77 $\pm$ 0.78	73.94 $\pm$ 1.06
NodeFormer	2022	64.49 $\pm$ 0.73	43.86 $\pm$ 0.35	86.71 $\pm$ 0.88	78.10 $\pm$ 1.03	74.27 $\pm$ 1.46
DIFFormer	2023	79.10 $\pm$ 0.32	47.84 $\pm$ 0.65	90.89 $\pm$ 0.58	83.57 $\pm$ 0.68	72.15 $\pm$ 1.31
GOAT	2023	71.59 $\pm$ 1.25	44.61 $\pm$ 0.50	81.09 $\pm$ 1.02	83.11 $\pm$ 1.04	75.76 $\pm$ 1.66
Polynormer	2024	92.13 $\pm$ 0.50	54.46 $\pm$ 0.40	96.96 $\pm$ 0.52	84.83 $\pm$ 0.72	77.95 $\pm$ 1.06
Polynormer-r	2024	92.42 $\pm$ 0.36	54.54 $\pm$ 0.43	97.19 $\pm$ 0.42	85.57 $\pm$ 0.63	78.11 $\pm$ 1.07
LORS-Polynormer Ours		92.77$\pm$0.39	54.86$\pm$0.48	97.56$\pm$0.22	85.69$\pm$0.85	78.37$\pm$0.89

4.2 Effect on Advanced Graph Transformer (Tokenphormer)

To further evaluate the effectiveness of LORS, we integrate it into Tokenphormer, another state-of-the-art Graph Transformer model. The LORS integration applied to Tokenphormer is described in Section 3.3. Table 5 reports the results comparing LORS-Tokenphormer with other SOTA GNNs and GTs on four homophilic datasets. LORS-Tokenphormer achieves the best overall performance, and LORS consistently improves Tokenphormer across all experiments—most notably on Citeseer—showing that LORS enhances the model’s ability to flexibly capture structural and contextual information.

Table 4. Averaged node classification results over 10 runs on heterophilic datasets — Accuracy is reported for roman-empire and amazon-ratings, while ROC AUC is reported for minesweeper, tolokers, and questions. Each run is completed at 10% of the total epochs. Polynormer-r indicates Polynormer with *ReLU* activation, while LORS-Polynormer denotes the integration of LORS with Polynormer-r. The best results are highlighted in **bold**.

	Year	roman-empire	amazon-ratings	minesweeper	tolokers	questions
Polynormer-r	2024	90.63±0.46	45.75±1.86	92.38±0.83	76.96±1.30	65.05±2.25
LORS-Polynormer Ours		91.19±0.37	52.32±0.64	94.66±0.40	81.82±1.43	67.52±1.93

Table 5. Averaged node prediction results with 10 runs on homophilic datasets — Accuracy is reported for all datasets. ‘X’ and ‘-’ indicate unavailable numbers. The best results are in **bold**.

	Year	Cora	Citeseer	DBLP	Pubmed
GCN	2017	87.33±0.38	79.43±0.26	83.62±0.13	86.54±0.12
GAT	2017	86.29±0.53	80.13±0.62	84.19±0.19	86.32±0.16
GraphSAGE	2018	86.90±0.84	79.23±0.53	84.73±0.28	86.19±0.18
APPNP	2018	87.15±0.43	79.33±0.35	84.40±0.17	88.43±0.15
JKNet	2018	87.70±0.65	78.43±0.31	84.57±0.28	87.64±0.26
GRAND+	2022	85.8X±0.4X	75.6X±0.4X	-	88.64±0.09
GT	2020	71.84±0.62	67.38±0.76	81.04±0.27	88.79±0.12
Graphormer	2021	72.85±0.76	66.21±0.83	80.93±0.39	82.76±0.24
SAN	2021	74.02±1.01	70.64±0.97	83.11±0.32	88.22±0.15
Gophormer	2021	87.85±0.10	80.23±0.09	85.20±0.20	89.40±0.14
ANS-GT	2022	88.60±0.45	80.25±0.39	-	89.56±0.55
GraphGPS	2022	-	-	-	88.94±0.16
Expormer	2023	-	-	-	89.52±0.54
Gapformer	2023	87.37±0.76	76.21±1.47	85.50±0.43	88.98±0.46
NAGphormer	2023	90.56±0.39	80.02±0.80	84.62±0.13	89.60±0.14
Tokenphormer	2025	90.39±0.81	79.67±0.76	84.24±0.25	88.97±0.46
LORS-Tokenphormer Ours		91.19±0.62	80.74±0.47	84.62±0.22	89.38±0.22

4.3 Effect on Traditional Graph Neural Networks

To demonstrate that LORS can enhance not only GTs but also traditional GNNs, we integrate it into several representative GNN architectures [6]. As shown in Table 6, we evaluate these LORS-GNN variants on the ZINC regression dataset under 20-epoch and 100-epoch training schedules. Under 20 epochs, LORS consistently reduces MAE by 10–30%. Notably, for a 500K-parameter GCN, MAE drops from 0.7694 to 0.5238—a 32% improvement. As expected, the gain decreases under the 100-epoch setting, though most models still benefit from noticeable MAE reductions.

Table 6. Mean Absolute Error (MAE) on the ZINC dataset for a regression task, evaluated with 100K and 500K parameter settings under 20 and 100 epoch training schedules. The best results are in **bold**.

	Year	100K 20 epoch ↓	500K 20 epoch ↓	100K 100 epoch ↓
GMM	2017	0.7117	0.5151	0.3983
GCN	2017	0.6195	0.7694	0.4195
ResGCN	2018	0.5484	0.4958	0.4846
GAT	2018	0.7099	0.5317	0.4773
GIN	2019	0.5586	6.1887	0.3716
LORS-GMM	Ours	0.5241	0.4499	0.3873
LORS-GCN	Ours	0.5077	0.5238	0.4086
LORS-ResGCN	Ours	0.5083	0.4809	0.4244
LORS-GAT	Ours	0.5400	0.4940	0.4567
LORS-GIN	Ours	0.5085	2.3112	0.3952

This may be attributed to GIN’s limited capacity for rapid convergence at larger scales. However, with LORS integrated, GIN’s convergence improves dramatically, reducing the MAE from 6.1887 to 2.3112—a relative improvement of over 60%. These results further highlight the robustness and adaptability of LORS under constrained or challenging training conditions.

4.4 Early-Stage Convergence Comparison

To better illustrate the effect of LORS in accelerating convergence, Figure 3 presents the performance comparison between Polynormer and LORS-Polynormer at the 10% epoch mark across five heterophilic graph datasets. LORS-Polynormer reaches this baseline with significantly fewer epochs than Polynormer, with improvements ranging from 18% to 53%, further highlighting the effectiveness of LORS in accelerating model convergence.

5 CONCLUSION

This paper presents a novel Graph Transformer framework, LORS-GT. We instantiate this framework in two representative models, LORS-Polynormer and

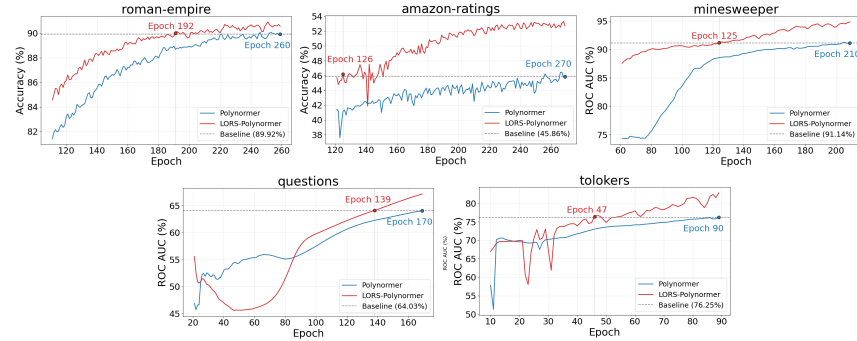


Figure 3. Convergence curve comparing *Polynormer* and *LORS-Polynormer* (heterophilic datasets)

LORS-Tokenphormer. Experimental results across diverse graph datasets indicate that our approach significantly accelerates convergence and enriches the representational capacity of Graph Transformers. However, its compatibility may vary across architectures. For instance, the integration framework LORS V3 may not be applicable to all models.

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